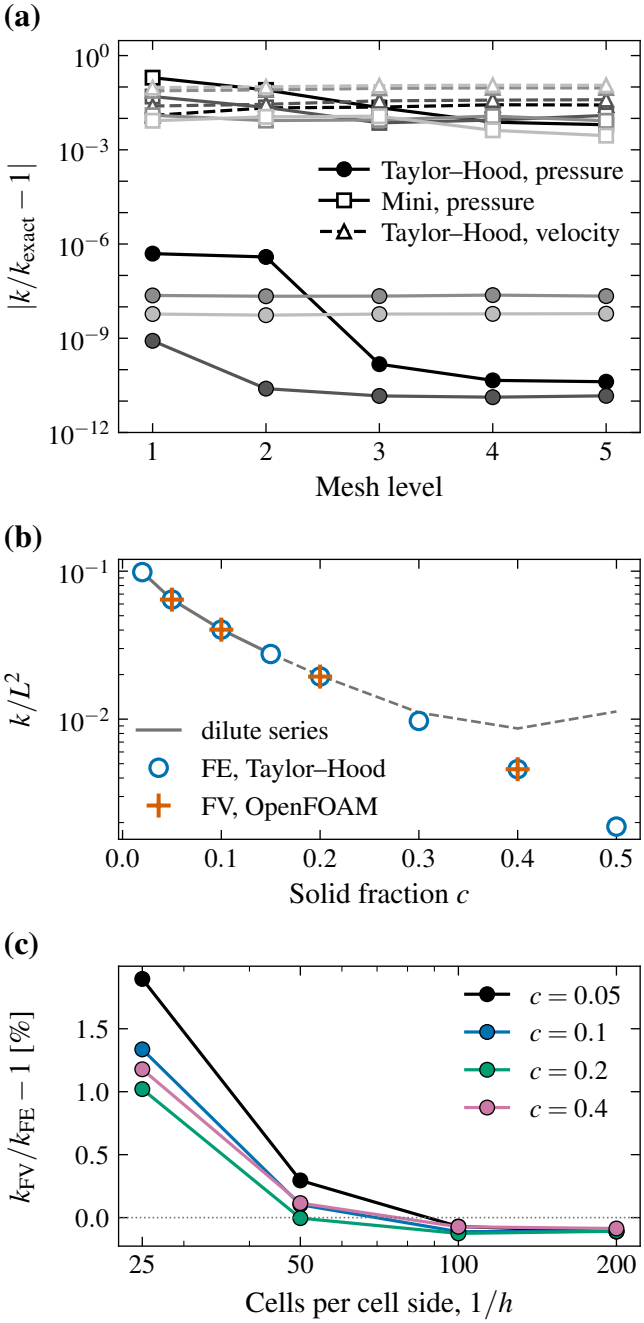

(a)
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
$|k/k_{\text{exact}} - 1|$
Taylor–Hood, pressure
Mini, pressure
Taylor–Hood, velocity
1
2
3
4
5
Mesh level
(b)
10^{-1}
10^{-2}
k/L^2
dilute series
FE, Taylor–Hood
FV, OpenFOAM
0.0
0.1
0.2
0.3
0.4
0.5
Solid fraction c
(c)
$k_{\text{FV}}/k_{\text{FE}} - 1$ [%]
1.5
1.0
0.5
0.0
$c = 0.05$
$c = 0.1$
$c = 0.2$
$c = 0.4$
25
50
100
200
Cells per cell side, $1/h$